

Name: Maximus 

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Objectives:

- (1.) I shall convert a measurement from a customary unit (Fluid Ounce) to a metric unit (Milliliters).
- (2.) I shall convert a measurement from a metric unit (Milliliters) to a customary unit (Fluid Ounce).

Measurement: Volume

First Unit: Customary Unit (Fluid Ounce)

To convert to: Metric Unit (Milliliter)

Second Unit: Metric Unit (Milliliter)

To convert to: Customary Unit (Fluid Ounce)

Container: Purell Hand Sanitizer Container



First Conversion: Convert 354 milliliters to fluid ounce

From the Tables that were given:

$$1 \text{ mL} = 10^{-3} L$$

$$1 \text{ mL} = 0.001 L$$

$$1 L = 0.26417205 \text{ gal}$$

$$1 \text{ gal} = 128 \text{ fl oz}$$

I shall be using the **Unity Fraction Method**

$$\dots \text{ mL} * \frac{\dots L}{\dots \text{ mL}} * \frac{\dots \text{ gal}}{\dots L} * \frac{\dots \text{ fl oz}}{\dots \text{ gal}}$$

$$354 \text{ mL} * \frac{0.001 L}{1 \text{ mL}} * \frac{0.26417205 \text{ gal}}{1 L} * \frac{128 \text{ fl oz}}{1 \text{ gal}}$$

$$= 11.97016393 \text{ fl oz}$$

$$\text{Value on the container} = 12 \text{ fl oz}$$

The calculated value was rounded to the nearest whole number to get the value on the container.

Second Conversion: Convert 12 fluid ounces to milliliters

From the Tables that were given:

$$1 \text{ mL} = 10^{-3} \text{ L}$$

$$1 \text{ mL} = 0.001 \text{ L}$$

$$1 \text{ L} = 0.26417205 \text{ gal}$$

$$1 \text{ gal} = 128 \text{ fl oz}$$

I shall be using the **Unity Fraction Method**

$$\dots \text{ fl oz} * \frac{\dots \text{ gal}}{\dots \text{ fl oz}} * \frac{\dots \text{ L}}{\dots \text{ gal}} * \frac{\dots \text{ mL}}{\dots \text{ L}}$$

$$12 \text{ fl oz} * \frac{1 \text{ gal}}{128 \text{ fl oz}} * \frac{1 \text{ L}}{0.26417205 \text{ gal}} * \frac{1 \text{ mL}}{0.001 \text{ L}}$$

$$= 354.8823579 \text{ mL}$$

$$\text{Value on the container} = 354 \text{ mL}$$

The calculated value was rounded down to the nearest whole number to get the value on the container.

Reference (MLA 9 Citation)

Chukwuemeka, Samuel. "Quantitative Reasoning Project: Measurements and Units on Containers." *Measurements and Units*,
conferencepresentations.appspot.com/Projects/ArithmeticAlgebra/MeasurementsUnits.html. Accessed 3 Oct. 2024.